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Method of providing information related to status of electric vehicle

Abstract

A system for providing information related to a charge status of an electric vehicle when the electric vehicle is connected to a charging station includes a system controller in communication with an electrical system within the electric vehicle and adapted to monitor information related to a charge status of the electric vehicle, and provide notification, to a third-party, of the charge status of the electric vehicle.

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Background/Summary

- (1) The present disclosure relates to systems and methods for providing information related to a charge status of an electric vehicle.
- (2) With increasing number of electric vehicles and the expanded use of electric vehicle, it is important that a driver of an electric vehicle can identify and evaluate potential charging stations. Currently, a driver must guess when deciding where to stop at a charging station location, or when selecting a specific charging station at which to queue. A system that provides notification to a third party of the charging status of an electric vehicle would allow a third party to strategically pick charging station locations and specific charging stations.
- (3) Thus, while current systems and methods achieve their intended purpose, there is a need for a new and improved system and method for providing information related to a charge status of an electric vehicle when the electric vehicle is connected to a charging station, whereby a third party can strategically select a charging station that minimizes the time spent waiting for such charging station.

SUMMARY

- (4) According to several aspects of the present disclosure a method of providing information related to a charge status of an electric vehicle when the electric vehicle is connected to a charging station, includes monitoring, with a system controller in communication with an electrical system within the electric vehicle, information related to a charge status of the electric vehicle, and providing notification, to a third-party, with the system controller, the charge status of the electric vehicle.
- (5) According to another aspect, the monitoring, with the system controller in communication with the electrical system within the electric vehicle, information related to a charge status of the electric vehicle further includes monitoring, with the system controller, at least one of: a level of charge of the electrical system within the electric vehicle, and a time left until the charging of the electric vehicle is complete.
- (6) According to another aspect, the monitoring, with the system controller, a time left until the charging of the electric vehicle is complete further includes calculating a time until one of: when the electric vehicle is fully charged, and when the electric vehicle is charged sufficiently to allow the electric vehicle to travel from the charging station to a planned destination.
- (7) According to another aspect, the providing notification, to a third-party, with the system controller, the charge status of the electric vehicle further includes: receiving, with the system controller of the electric vehicle, a charging status query from a third-party, wherein the third party initiates the charging status query based on at least one of: in response to manual initiation by the third party, automatically based on third party navigation and destination information, and automatically based on battery charge level of the third party and proximity to the charging station, and providing notification, to the third-party, with the system controller, the charge status of the electric vehicle in response to receiving the charging status query.
- (8) According to another aspect, the providing notification, to a third party, with the system controller, the charge status of the electric vehicle further includes providing a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, and receiving, by the third party, the visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.
- (9) According to another aspect, the providing a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete further includes, displaying, with an internal/external display system in communication with the system controller and adapted to project information onto a window surface of the vehicle, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, and a QR Code including the

current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

(10) According to another aspect, the providing a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete further includes displaying, with the system controller, via a display of the electric vehicle, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

(11) According to another aspect, the providing a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete further includes displaying, with the system controller, via a display screen of the charging station, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

(12) According to another aspect, the providing notification, to a third party, with the system controller, the charge status of the electric vehicle further includes: uploading, with the system controller, the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete to a cloud-based charging station network, accessing, by the third party, the cloud-based charging station network, and receiving, by the third party, from the cloud-based charging station network, the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

(13) According to another aspect, the providing notification, to a third party, with the system controller, the charge status of the electric vehicle further includes sending, with the system controller, notification to the third party that the charging station to which the electric vehicle is connected has malfunctioned.

(14) According to several aspects of the present disclosure, a system for providing information related to a charge status of an electric vehicle when the electric vehicle is connected to a charging station includes a system controller in communication with an electrical system within the electric vehicle and adapted to: monitor information related to a charge status of the electric vehicle, and provide notification, to a third-party, of the charge status of the electric vehicle.

(15) According to another aspect, when monitoring information related to a charge status of the electric vehicle, the system controller is further adapted to monitor a level of charge of the electrical system within the electric vehicle, and a time left until the charging of the electric vehicle is complete.

(16) According to another aspect, when monitoring a time left until the charging of the electric vehicle is complete the system controller is further adapted to calculate at least one of: a time until the electric vehicle is fully charged, and a time until the vehicle is charged sufficiently to allow the electric vehicle to travel from the charging station to a planned destination.

(17) According to another aspect, when providing notification, to a third-party, the charge status of the electric vehicle, the system controller is further adapted to: receive a charging status query from a third-party, wherein the third party initiates the charging status query based on at least one of: in response to manual initiation by the third party, automatically, based on third party navigation and destination information, and automatically, based on battery charge level of the third party and proximity to the charging station, and provide notification, to the third-party, the charge status of the electric vehicle in response to receiving the charging status query.

(18) According to another aspect, when providing notification, to a third party, the charge status of

the electric vehicle, the system controller is further adapted to provide a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, wherein, the third party receives the visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

(19) According to another aspect, when providing a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, the system controller is further adapted to display, with an internal/external display system in communication with the system controller and adapted to project information onto a window surface of the vehicle, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

(20) According to another aspect, when providing a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, the system controller is further adapted to display, with a display of the electric vehicle, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

(21) According to another aspect, when providing a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, the system controller is further adapted to display, with a display screen of the charging station, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

(22) According to another aspect, when providing notification, to a third party, the charge status of the electric vehicle, the system controller is further adapted to: upload the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete to a cloud-based charging station network, wherein the cloud-based charging station is adapted to allow access by the third party, and the third party receives the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete from the cloud-based charging station network, and send notification to the third party that the charging station to which the electric vehicle is connected has malfunctioned.

(23) Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

(2) FIG. 1 is a schematic diagram of a vehicle having a system in accordance with an exemplary embodiment of the present disclosure;

(3) FIG. 2 is a schematic diagram of a system according to an exemplary embodiment;
(4) FIG. 3 is a schematic diagram of a third-party vehicle receiving notifications from several electric vehicles via a cloud-based charging station network;
(5) FIG. 4A is a schematic front view of a vehicle that is connected to a charging station;
(6) FIG. 4B is a schematic side view of the vehicle shown in FIG. 4A;
(7) FIG. 4C is schematic front view of a vehicle that is connected to a charging station and is providing notification by way of a QR Code displayed on the front windshield;
(8) FIG. 5 is a schematic view of a hybrid augmented reality head up display system for the system;
(9) FIG. 6 is schematic illustration of a substrate for the windshield of the electric vehicle according to an exemplary embodiment;
(10) FIG. 7 is a enlarged portion of FIG. 6, as indicated by the circle labelled “7” in FIG. 6; and
(11) FIG. 8 is a flow chart illustrating a method according to an exemplary embodiment.
(12) The figures are not necessarily to scale and some features may be exaggerated or minimized, such as to show details of particular components. In some instances, well-known components, systems, materials or methods have not been described in detail in order to avoid obscuring the present disclosure. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

DETAILED DESCRIPTION

(13) The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features. As used herein, the term module refers to any hardware, software, firmware, electronic control component, processing logic, and/or processor device, individually or in any combination, including without limitation: application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality. Although the figures shown herein depict an example with certain arrangements of elements, additional intervening elements, devices, features, or components may be present in actual embodiments. It should also be understood that the figures are merely illustrative and may not be drawn to scale.

(14) As used herein, the term “vehicle” is not limited to automobiles. While the present technology is described primarily herein in connection with automobiles, the technology is not limited to automobiles. The concepts can be used in a wide variety of applications, such as in connection with aircraft, marine craft, other vehicles, and consumer electronic components.

(15) In accordance with an exemplary embodiment, FIG. 1 shows an electric vehicle **10** with an associated system **11** for providing information related to a charge status of the electric vehicle **10** when the electric vehicle is connected to a charging station in accordance with various embodiments. In general, the system **11** works in conjunction with other systems within the vehicle **10** to display various information and infotainment content for the passenger. The vehicle **10** generally includes a chassis **12**, a body **14**, front wheels **16**, and rear wheels **18**. The body **14** is arranged on the chassis **12** and substantially encloses components of the vehicle **10**. The body **14** and the chassis **12** may jointly form a frame. The front wheels **16** and rear wheels **18** are each rotationally coupled to the chassis **12** near a respective corner of the body **14**.

(16) In various embodiments, the vehicle **10** is an autonomous vehicle and the system **11** is incorporated into the autonomous vehicle **10**. An autonomous vehicle **10** is, for example, a vehicle **10** that is automatically controlled to carry passengers from one location to another. The vehicle **10** is depicted in the illustrated embodiment as a passenger car, but it should be appreciated that any

other vehicle including motorcycles, trucks, sport utility vehicles (SUVs), recreational vehicles (RVs), etc., can also be used. In an exemplary embodiment, the vehicle **10** is equipped with a so-called Level Four or Level Five automation system. A Level Four system indicates “high automation”, referring to the driving mode-specific performance by an automated driving system of all aspects of the dynamic driving task, even if a human driver does not respond appropriately to a request to intervene. A Level Five system indicates “full automation”, referring to the full-time performance by an automated driving system of all aspects of the dynamic driving task under all roadway and environmental conditions that can be managed by a human driver.

(17) As shown, the vehicle **10** generally includes a propulsion system **20**, a transmission system **22**, a steering system **24**, a brake system **26**, a sensor system **28**, an actuator system **30**, at least one data storage device **32**, a vehicle controller **34**, and a communication system **36**. In an embodiment in which the vehicle **10** is an electric vehicle, there may be no transmission system **22**. The propulsion system **20** may, in various embodiments, include an internal combustion engine, an electric machine such as a traction motor, and/or a fuel cell propulsion system. The transmission system **22** is configured to transmit power from the propulsion system **20** to the vehicle's front wheels **16** and rear wheels **18** according to selectable speed ratios. According to various embodiments, the transmission system **22** may include a step-ratio automatic transmission, a continuously-variable transmission, or other appropriate transmission. The brake system **26** is configured to provide braking torque to the vehicle's front wheels **16** and rear wheels **18**. The brake system **26** may, in various embodiments, include friction brakes, brake by wire, a regenerative braking system such as an electric machine, and/or other appropriate braking systems. The steering system **24** influences a position of the front wheels **16** and rear wheels **18**. While depicted as including a steering wheel for illustrative purposes, in some embodiments contemplated within the scope of the present disclosure, the steering system **24** may not include a steering wheel.

(18) The sensor system **28** includes one or more sensing devices **40a-40n** that sense observable conditions of the exterior environment and/or the interior environment of the autonomous vehicle **10**. The sensing devices **40a-40n** can include, but are not limited to, radars, lidars, global positioning systems, optical cameras, thermal cameras, ultrasonic sensors, and/or other sensors. The cameras can include two or more digital cameras spaced at a selected distance from each other, in which the two or more digital cameras are used to obtain stereoscopic images of the surrounding environment in order to obtain a three-dimensional image. The plurality of sensing devices **40a-40n** is used to determine information about an environment surrounding the vehicle **10**. In an exemplary embodiment, the plurality of sensing devices **40a-40n** includes at least one of a motor speed sensor, a motor torque sensor, an electric drive motor voltage and/or current sensor, an accelerator pedal position sensor, a coolant temperature sensor, a cooling fan speed sensor, and a transmission oil temperature sensor. In another exemplary embodiment, the plurality of sensing devices **40a-40n** further includes sensors to determine information about the environment surrounding the vehicle **10**, for example, an ambient air temperature sensor, a barometric pressure sensor, and/or a photo and/or video camera which is positioned to view the environment in front of the vehicle **10**. In another exemplary embodiment, at least one of the plurality of sensing devices **40a-40n** is capable of measuring distances in the environment surrounding the vehicle **10**.

(19) In a non-limiting example wherein the plurality of sensing devices **40a-40n** includes a camera, the plurality of sensing devices **40a-40n** measures distances using an image processing algorithm configured to process images from the camera and determine distances between objects. In another non-limiting example, the plurality of vehicle sensors **40a-40n** includes a stereoscopic camera having distance measurement capabilities. In one example, at least one of the plurality of sensing devices **40a-40n** is affixed inside of the vehicle **10**, for example, in a headliner of the vehicle **10**, having a view through the windshield **56** of the vehicle **10**. In another example, at least one of the plurality of sensing devices **40a-40n** is affixed outside of the vehicle **10**, for example, on a roof of the vehicle **10**, having a view of the environment surrounding the vehicle **10**. It should be

understood that various additional types of sensing devices, such as, for example, LiDAR sensors, ultrasonic ranging sensors, radar sensors, and/or time-of-flight sensors are within the scope of the present disclosure. The actuator system **30** includes one or more actuator devices **42a-42n** that control one or more vehicle **10** features such as, but not limited to, the propulsion system **20**, the transmission system **22**, the steering system **24**, and the brake system **26**.

(20) The vehicle controller **34** includes at least one processor **44** and a computer readable storage device or media **46**. The at least one data processor **44** can be any custom made or commercially available processor, a central processing unit (CPU), a graphics processing unit (GPU), an auxiliary processor among several processors associated with the vehicle controller **34**, a semi-conductor based microprocessor (in the form of a microchip or chip set), a macro-processor, any combination thereof, or generally any device for executing instructions. The computer readable storage device or media **46** may include volatile and nonvolatile storage in read-only memory (ROM), random-access memory (RAM), and keep-alive memory (KAM), for example. KAM is a persistent or non-volatile memory that may be used to store various operating variables while the at least one data processor **44** is powered down. The computer-readable storage device or media **46** may be implemented using any of a number of known memory devices such as PROMs (programmable read-only memory), EPROMs (electrically PROM), EEPROMs (electrically erasable PROM), flash memory, or any other electric, magnetic, optical, or combination memory devices capable of storing data, some of which represent executable instructions, used by the controller **34** in controlling the vehicle **10**.

(21) The instructions may include one or more separate programs, each of which includes an ordered listing of executable instructions for implementing logical functions. The instructions, when executed by the at least one processor **44**, receive and process signals from the sensor system **28**, perform logic, calculations, methods and/or algorithms for automatically controlling the components of the vehicle **10**, and generate control signals to the actuator system **30** to automatically control the components of the vehicle **10** based on the logic, calculations, methods, and/or algorithms. Although only one controller **34** is shown in FIG. **1**, embodiments of the vehicle **10** can include any number of controllers **34** that communicate over any suitable communication medium or a combination of communication mediums and that cooperate to process the sensor signals, perform logic, calculations, methods, and/or algorithms, and generate control signals to automatically control features of the autonomous vehicle **10**.

(22) In various embodiments, one or more instructions of the vehicle controller **34** are embodied in a trajectory planning system and, when executed by the at least one data processor **44**, generates a trajectory output that addresses kinematic and dynamic constraints of the environment. For example, the instructions receive as input process sensor and map data. The instructions perform a graph-based approach with a customized cost function to handle different road scenarios in both urban and highway roads.

(23) The communication system **36** is configured to wirelessly communicate information to and from other remote entities **48**, such as but not limited to, other vehicles ("V2V" communication) infrastructure ("V2I" communication), remote systems, remote servers, cloud computers, and/or personal devices. In an exemplary embodiment, the communication system **36** is a wireless communication system configured to communicate via a wireless local area network (WLAN) using IEEE 802.11 standards or by using cellular data communication. However, additional or alternate communication methods, such as a dedicated short-range communications (DSRC) channel, are also considered within the scope of the present disclosure. DSRC channels refer to one-way or two-way short-range to medium-range wireless communication channels specifically designed for automotive use and a corresponding set of protocols and standards.

(24) The vehicle controller **34** is a non-generalized, electronic control device having a preprogrammed digital computer or processor, memory or non-transitory computer readable medium used to store data such as control logic, software applications, instructions, computer code,

data, lookup tables, etc., and a transceiver [or input/output ports]. Computer readable medium includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A “non-transitory” computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device. Computer code includes any type of program code, including source code, object code, and executable code.

(25) Referring to FIG. 2, the system **11** is shown in more detail, wherein, the system **11** includes a system controller **34A**. The system controller **34A** may be the vehicle controller **34**, or the system controller **34A** may be a separate controller in communication with the vehicle controller **34**. The system controller **34A** is in communication, over a vehicle communication bus **50**, with an electrical system **52** of the electric vehicle **10**. The electrical system **52** includes a battery pack **54**, power inverter **56**, electric motor **58**, onboard battery charger **60** and a battery management system **62**.

(26) The system controller **34A** is adapted to monitor information related to a charge status of the electric vehicle **10** when the electric vehicle **10** is connected to a charging station **64**. In an exemplary embodiment, the system controller **34A** is adapted to monitor a level of charge of the electrical system **52** within the electric vehicle **10** and a time left until the charging of the electric vehicle **10** is complete.

(27) In an exemplary embodiment, the system controller **34A** is adapted to calculate a pre-determined threshold based on at least one of a time until the electric vehicle **10** is fully charged and a time until the level of charge of the electric vehicle **10** is sufficient to allow the electric vehicle **10** to travel from a current location to a planned destination. For example, the pre-determined threshold may be based on when the electric vehicle **10** is fully charged, wherein, when the electric vehicle **10** is plugged into a charging station **64**, the system controller **34A** will monitor the charging of the electric vehicle **10** and the time left until the electric vehicle **10** is charged to the pre-determined threshold will be the time left until the electric vehicle **10** is fully charged. In another example, the electric vehicle **10** may be located at a charging station **64** that is one hundred miles from the owner's home. In order to ensure that the owner of the electric vehicle **10** spends as little time as possible at the charging station **64**, the pre-determined threshold is based on when the electric vehicle **10** is charged to a sufficient level to allow the electric vehicle **10** to travel from the charging station **64** to the owner's home, or the next charging station **64**. Possibly, the electric vehicle **10** will only need to be charged to 35% or more to ensure that the electric vehicle **10** will be able to travel the one hundred miles to the owner's home, wherein the owner can plug the electric vehicle **10** into a home charger to fully charge the electric vehicle **10**.

(28) When calculating the pre-determined threshold, the system controller **34A** receives information related to the planned destination from at least one of input from a user of the electric vehicle **10**, and input from a navigation system within the electric vehicle **10**, wherein the planned destination is one of a charging station or a home location. The system controller **34A** receives information for the planned destination, which is the destination of the electric vehicle **10** after the electric vehicle **10** leaves a current location. Information about the planned destination may be entered directly, or the system controller **34A** may gather such information by accessing a navigation system or a personal device of a driver of the electric vehicle **10**.

(29) In another example, the system controller **34A** receives information for the planned destination, even when a planned destination has not been indicated. The system controller, via communication with electric vehicle charging station infrastructure networks can identify how far it is to the next available charging station **64** and designate that charging station **64** as the planned destination. In this example, the system controller **34A** receives information that the electric vehicle

10 will need to travel at least one hundred miles from the current location to the planned destination (closest available charging station). The electric vehicle **10** will need at least a 25% charge to travel one hundred miles, so the system controller **34A** sets the pre-determined threshold at 25%. Thus, the system controller **34A** calculates the time remaining until charging of the electric vehicle **10** is complete by calculating the time it will take for the electric vehicle **10** to be charged up to the pre-determined threshold.

(30) In another exemplary embodiment, the pre-determined threshold is automatically calculated with a safety margin to accommodate for unexpected stops or detours. For example, if the electric vehicle **10** needs 25% battery charge to travel one hundred miles to the person's **66** home, the system controller **34A** will calculate the pre-determined threshold as 25%, the minimum battery charge necessary to travel the one hundred miles, plus an additional 10% to account for an unexpected stop or a detour due to construction or traffic, etc. Thus, the pre-determined threshold in this example is calculated at 35%.

(31) In another exemplary embodiment, when calculating the pre-determined threshold, the system controller **34A** receives information related to the planned destination from stored history of previous trip patterns. For example, if an electric vehicle **10** has been located at a particular location regularly over a period of time, such as six months, and over that six-month time frame, each day, the electric vehicle **10** travels, first, from that location to a convenience store, then to a day care center to pick up a child, and then to the driver's home. The system controller **34A** may recognize this pattern, and determine the planned destination (convenience store, day care center, then home) and calculate the pre-determined threshold based on assuming that pattern will be repeated, even when no direct information is provided on what the planned destination is.

(32) The system controller **34A** is further adapted to provide notification, to a third party **66**, information related to a charge status of the electric vehicle **10**, including, at least one of (1) a level of charge of the electrical system **52** within the electric vehicle **10**, and (2) a time left until charging of the electric vehicle **10** is complete. The third party **66** may be a person **66A**, or a third-party electric vehicle **66B**. In an exemplary embodiment, the system controller **34A** is adapted to receive a charging status query from the third-party **66**. The third party **66** initiates the charging status query based on at least one of (1) in response to manual initiation by the third party **66**, (2) automatically, based on third party navigation and destination information, and (3) automatically, based on battery charge level of a third-party vehicle **66B** and proximity to the charging station **64**, wherein, the system controller **34A** provides notification, to the third-party **66**, the charge status of the electric vehicle **10** in response to receiving the charging status query.

(33) Referring to FIG. 3, for example, the third party **66** may manually initiate a charging status query, wherein the third party **66** is a driver **66A** of a third-party electric vehicle **66B**, notices that the level of charge of the third-party electric vehicle **66B** is getting low and also notices that the third-party electric vehicle **66B** is approaching a charging station **64**. The third party driver **66A** may initiate a charging status query to find out how long they may have to wait for availability of the upcoming charging station **64**. As shown, the third-party electric vehicle **66B** is approaching three potential charging stations **64A**, **64B**, **64C**. The three potential charging stations **64A**, **64B**, **64C** may be individual charging stations at a specific location, or may be individual charging stations located at different locations. The third party driver **66A** initiates a charging status query, wherein the system controller **34A**, via wireless communication with a charging station network **65**, identifies the three potential charging stations **64A**, **64B**, **64C**.

(34) In an exemplary embodiment, the charging station network **65** is a cloud-based application or network service that communicates with all charging stations **64A**, **64B**, **64C** to manage availability, appropriateness for an electric vehicle **10**, and functionality of charging stations **64A**, **64B**, **64C**. In another example, the onboard systems of the third-party electric vehicle **66B** detect that the third-party electric vehicle **66B** will need to re-charge before reaching its' final destination, based on the battery charge level of the third-party electric vehicle **66B** and communication with

the third-party electric vehicle's **66B** navigation systems. Further, the onboard systems of the third-party electric vehicle **66B** detect that the third-party electric vehicle **66B** is approaching the three potential charging stations **64A**, **64B**, **64C**, based on wireless communications with the charging station network **65**. The onboard systems of the third-party electric vehicle **66B** will automatically initiate a charging status query to evaluate the availability of the three potential charging stations **64A**, **64B**, **64C** based on the charging status of three electric vehicles **10A**, **10B**, **10C** connected to them. For each of the electric vehicles **10A**, **10B**, **10C**, the system controller **34A** is adapted to upload the current level of charge and the time remaining until charging is complete to the cloud-based charging station network **65**. The cloud-based charging station network **65** is adapted to allow access by authorized third parties **66**. The onboard systems of the third-party electric vehicle **66B** may then access the cloud-based charging station network **65**, receive data related to the charging status of the electric vehicles **10A**, **10B**, **10C**, and compare availability (vacant/occupied, time until charging of present electric vehicle **10A**, **10B**, **10C** is complete, functional, appropriate for the third-party electric vehicle **66B**) of the potential charging stations **64A**, **64B**, **64C** to either automatically select the best option for charging the third-party electric vehicle **66B** (ie. autonomous third-party electric vehicles) or to provide suggestions to the driver **66A** of the third-party electric vehicle **66B**. Further, if a charging station **64** has malfunctioned, the cloud-based charging station network **65** can provide notification to the third party **66** that a charging station **64** to which the electric vehicle **10** is connected has malfunctioned, thus eliminating that charging station **64** as an option for the third party **66**.

(35) In an exemplary embodiment, when providing notification **72**, to a third party **66**, the charge status of the electric vehicle **10**, the system controller **34A** is further adapted to provide a visual representation **72A** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, wherein, the third party **66** receives the visual representation **72A** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete.

(36) In an exemplary embodiment, when displaying the visual representation **72A** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, the system controller **34A** is further adapted to display, with an internal/external display system **68** in communication with the system controller **34A** and adapted to project information onto a window surface **70** of the electric vehicle **10**, information related to the charge status of the electric vehicle **10**, wherein, the projected information related to the charge status of the electric vehicle **10** is visible from within the electric vehicle **10** and from outside the electric vehicle **10**. The system controller **34A** is adapted to project information including at least one of: (1) the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, (2) a graphic representation **72B** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, and (3) a QR Code **72C** including the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete.

(37) In an exemplary embodiment, the internal/external display system **68** comprises a hybrid augmented reality head up display (HUD) **68**. The system controller **34A** uses the HUD **68** to provide a notification **72** of the charge status of the electric vehicle **10** onto the windshield **70A**. As shown in FIG. **4A**, the system controller **34A** uses the internal/external display system **68** to display a graphic representation **72B**. As shown, the graphic representation **72B** comprises a battery icon with bars indicating a relative level of charge, with one bar being highlighted, thus indicating that the electric vehicle **10** is approximately 20% charged. While, as shown, the graphic representation **72B** does not provide a specific time until charging of the electric vehicle is complete, the graphic representation **72B** does provide information to allow a driver **66A** of a third-party electric vehicle **66B** to visually compare different electric vehicles **10** at different charging stations **64**. Thus, as a third-party electric vehicle **66B** approaches various electric vehicles **10A**, **10B**, **10C** connected to

various charging stations **64A**, **64B**, **64C** (as shown in FIG. 3), the driver of the third-party electric vehicle **66B** can look at the displayed notification **72** to evaluate the status of the different charging stations **64A**, **64B**, **64C** and decide which one is the best option, thereafter, entering a queue for the selected charging station **64**. Further, while waiting, the driver **66A** of the third-party electric vehicle **66B** can look at the displayed notification **72** on electric vehicles **10** parked at nearby or adjacent charging stations **64** to continually monitor their charging status, and if appropriate, move to a different charging station **64** if the waiting time becomes less than the charging station **64** where the third-party vehicle **66B** is currently in line.

(38) Likewise, referring to FIG. 4B, in another exemplary embodiment, the system controller **34A** uses the HUD **68** to display the current charge level **74A** of the electric vehicle **10** and the calculated time **74B** until charging of the electric vehicle **10** is complete onto a side window **70B**. The displayed current charge level **74A** and time remaining **74B** may be represented with textual information, such as shown in FIG. 4B, or may be shown with a graphic representation **72B** that provide an illustrative representation of the information, as shown in FIG. 4A.

(39) Referring to FIG. 4C, in another exemplary embodiment, the system controller **34A** uses the HUD **68** to display a QR Code **72C** including the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete onto either the windshield **70A** or the side window **70B**. The QR code **72C** provides a visual representation **72A** of information that can be read by cameras on a third-party vehicle **66B** from a distance greater than could be visually identified by a human third party **66A**. Further, a displayed QR Code **72C** provides the information in a manner that can only be recognized by third party vehicles **66B** that are authorized to get the information, such as third party vehicles **66B** that have submitted a query through an authorized charging station network **65**.

(40) Thus, as a third-party electric vehicle **66B** approaches various electric vehicles **10A**, **10B**, **10C** connected to various charging stations **64A**, **64B**, **64C**, cameras within the third-party vehicle **66C** capture images of QR Codes **72C** displayed by electric vehicles **10A**, **10B**, **10C** connected to the various charging stations **64A**, **64B**, **64C** to evaluate the status of the different charging stations **64A**, **64B**, **64C** and decide which one is the best option, wherein, the onboard systems of the third-party vehicle **66B** can recommend a charging station **64** to the driver **66A** of the third-party vehicle **66B**, or, in the case of an autonomous vehicle, automatically select and enter a queue for the selected charging station **64**. Further, while waiting, the onboard systems of the third-party vehicle **66B** can monitor displayed QR Codes **72C** on electric vehicles **10** parked at nearby or adjacent charging stations **64** to continually monitor their charging status, and if appropriate, move to a different charging station **64** if the waiting time becomes less than the charging station **64** where the third-party vehicle **66B** is currently in line.

(41) In another exemplary embodiment, when providing a visual representation **72A** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, the system controller **34A** is further adapted to display, with a display **76** of the electric vehicle **10**, at least one of (1) the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, (2) a graphic representation **72B** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, and (3) a QR Code **72C** including the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete.

(42) Referring again to FIG. 4A, a graphical representation **72B** of a charging battery is displayed on a display **76** mounted to the electric vehicle **10**. As shown, the graphical representation **72B** displayed with the display **76** is the same graphical representation **72B** displayed on the front windshield **70A**. As shown, the display **76** is incorporated into a headlight assembly of the electric vehicle **10**. In such instance, the headlight may be an LED light array, or an OLED display, adapted to display information related to the current charge **74A** of the electric vehicle **10** and/or a time left

74B until charging of the electric vehicle **10** is complete. Referring again to FIG. 4C, a QR Code **72C** is displayed on the display **76**. It should be understood that such display **76** may be incorporated into a headlight assembly (as shown in FIG. 4A and FIG. 4C) or tail-light assembly of the electric vehicle **10**, or may be mounted independently onto an exterior surface of the electric vehicle **10**, separate from either the headlight assembly or tail-light assembly.

(43) In another exemplary embodiment, when providing a visual representation **72A** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, the system controller **34A** is further adapted to display, via communication with a connected charging station **64**, with a display screen **77** of the charging station **64**, at least one of (1) the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, (2) a graphic representation **72B** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, and (3) a QR Code **72C** including the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete.

(44) Referring again to FIG. 4A, textual data including the current charge level **74A** of the electric vehicle **10** and the time **74B** until the charging of the electric vehicle **10** is complete is displayed on the display screen **77** of the charging station **64**. Referring again to FIG. 4C, a QR Code **72C** is displayed on the display screen **77** of the charging station **64**.

(45) Thus, as shown in FIG. 4A, FIG. 4B and FIG. 4C, according to the examples provided above, the system controller **34A** provides a visual representation **72A** of a notification **72** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete.

(46) Displaying information such as the level of charge **74A** of the electric vehicle **10**, and the time remaining **74B** until charging of the electric vehicle **10** is complete with the internal/external display system **68** such as the HUD **68**, allows a third party (either driver **66A** of a third-party vehicle **66B** or onboard systems of a third-party vehicle **66B**) to see the displayed information, and use the displayed information to make strategic decisions regarding when to charge the third-party vehicle **66B** and selection of an appropriate charging station **64**.

(47) The internal/external display system **68** may use any suitable transparent display technology to display information onto the window surfaces **70** of the vehicle **10** to be viewed by persons both internally and externally, or, to be viewed only by persons external to the vehicle **10**. Referring to FIG. 5, in an exemplary embodiment, the HUD **68** includes a digital light processing (DLP) projector **78** adapted to project images **80** onto a window surface **70** of the vehicle **10**. The DLP **78** projects an image **80** that may contain various information, such as, by way of example, the level of charge **74**, **74A**, time remaining **76**, warning message **76A** and/or graphics **76B**, as shown in FIG. 3, FIG. 4 and FIG. 5. The DLP **78** is capable of projecting two-dimensional and three-dimensional images. In an exemplary embodiment, the DLP **78** includes a light source **82** adapted to project an excitation light **84**, a condensing lens **86** adapted to focus the excitation light **84** from the light source **82**, a color filter **88** (color wheel) adapted to split the focused excitation light **84** into red, green and blue light, a shaping lens **90** adapted to focus the excitation light **84** passing through the color filter **88**, a digital micro-mirror device (DMD) **92** adapted to re-direct the excitation light **84**, and a projection lens **94** adapted to receive the excitation light **84** from the digital micro-mirror device (DMD) **92** and project the excitation light **84** to the window surface **70**.

(48) Referring to FIG. 2, the system controller **34A** includes a DLP engine **96**. The DLP engine **96** includes a programmable processor including programming to monitor various inputs and determine what information is appropriate to display. The DLP engine **96** can communicate directly with various systems and components, or the DLP engine **96** can alternatively or additionally communicate over a LAN/CAN system. The window surfaces **70**, **70A**, **70B** are equipped with features capable of displaying an image projected thereupon while remaining transparent or substantially transparent such that occupants within the vehicle **10** can clearly see

displayed images **80** and still observe outside of the electric vehicle **10** through the window surfaces **70**, **70A**, **70B**, and persons **66A** outside of the electric vehicle **10** can clearly see displayed images **80** and still observe within of the electric vehicle **10** through the window surfaces **70**, **70A**, **70B**.

(49) The DLP engine **96** includes display software or programming translating requests to display information from the DLP engine **96** in graphical representations describing the information. The DLP engine **96** includes programming to compensate for the curved and tilted surface of the window surfaces **70**, **70A**, **70B**. The DLP engine **96** controls the light source **82** which includes a laser or projector device producing an excitation light **84** to project the images **80**.

(50) The window surfaces **70**, **70A**, **70B** function as a medium through which relevant features are observable while serving as a display device upon which images **80**, such as the current charge **74A** of the electric vehicle **10** and the time remaining **74B** until charging of the electric vehicle **10** is complete, as shown in FIG. **4A**, FIG. **4B** and FIG. **4C**, may be displayed. The window surfaces **70**, **70A**, **70B** are both transparent and capable of displaying images projected by an excitation light **84**. Referring to FIG. **6** and FIG. **7**, a person **66A** is able to see an arbitrary object (e.g. cube **98**) through a substrate **100** positioned on the window surfaces **70**, **70A**, **70B**. The substrate **100** may be transparent or substantially transparent. While the person **66A** sees the arbitrary object **98** through the substrate **100**, the person **66A** can also see images (e.g. circle **102** and triangle **104**) that are created at the substrate **100**. The substrate **100** may be part of the window surfaces **70**, **70A**, **70B**, a glass substrate, a plastic substrate, a polymer substrate, or other transparent (or substantially transparent) medium that would be appreciated by one of ordinary skill in the art. Other substrates may complement the substrate **100** to provide for tinting, substrate protection, light filtering (e.g. filtering external ultraviolet light), and other functions.

(51) FIG. **8** depicts illumination of transparent displays illuminated with excitation light **84** (e.g. ultraviolet light from a light source **82**). The substrate **100** receives excitation light **84** from the light source **82**. The received excitation light **84** may be absorbed by light emitting material **106** at the substrate **100**. When the light emitting material receives the excitation light **84**, the light emitting material **106** emits visible light. Accordingly, images (e.g. circle **102** and triangle **104**) may be created at the substrate **100** by selectively illuminating the substrate **100** with excitation light **84**.

(52) In an exemplary embodiment, the light emitting material **106** includes transparent phosphors that are embedded into the substrate **100**. The transparent phosphors are light emitting particles which fluoresce in response to being excited by the excitation light **84**. In an exemplary embodiment, the transparent phosphors are red, green, and blue (RGB) phosphors, allowing full color. The use of monochrome and/or two-color phosphors is also within the scope of the present disclosure. When excitation light **84** is absorbed by the transparent phosphors, visible light is emitted by the transparent phosphors. The excitation light **84** is provided by the light source **82**. Use of the substrate **100** and light emitting material **106** to display graphics is discussed in greater detail in U.S. application Ser. No. 17/749,464 titled "HYBRID AUGMENTED REALITY HEAD-UP DISPLAY FOR CREATING AN EDGE-TO-EDGE AUGMENTED REALITY VIEW" filed on May 20, 2022, the entire contents of which is hereby incorporated by reference.

(53) The excitation light **84** may be ultraviolet light, in accordance with embodiments of the present disclosure. If the excitation light **84** is ultraviolet light, then when the light emitting material **90** emits visible light in response to the ultraviolet light, a down-conversion physical phenomenon occurs. Specifically, ultraviolet light has a shorter wavelength and higher energy than visible light. Accordingly, when the light emitting material **106** absorbs the ultraviolet light and emits lower energy visible light, the ultraviolet light is down-converted to visible light because the ultraviolet light's energy level goes down when it is converted into visible light.

(54) As shown in FIG. **5**, the excitation light **84** is output by the light source **82** of the DLP **78**. In an exemplary embodiment, the light source **82** is a micro-mirror array (MMA) projector (e.g. a digital light processing (DLP) projector **78**). In the DLP **78**, the images **80** are created by microscopically small mirrors laid out in a matrix on a semiconductor chip within the DMD **92**. An

MMA projector that outputs ultraviolet light may be similar to a MMA projector that outputs visible light, except that the color filter **68** has light filters that are tailored to the ultraviolet light spectrum. In other embodiments, the DLP **78** is a liquid crystal display (LCD) projector. In embodiments, the DLP **78** may be a liquid crystal on silicon (LCOS) projector. In embodiments, the DLP **78** may be an analog projector (e.g. a slide film projector or a movie film projector). One of ordinary skill in the art would appreciate other types of projectors which may be used to project ultraviolet light on the substrate **100**.

(55) Referring to FIG. **8**, a method **200** of providing information related to a charge status of an electric vehicle **10** when the electric vehicle **10** is connected to a charging station **64** includes, beginning at block **202**, and moving to block **204**, monitoring, with a system controller **34A** in communication with an electrical system **52** within the electric vehicle **10**, information related to a charge status of the electric vehicle **10**, and, moving to block **206**, providing notification, to a third-party **66**, with the system controller **34A**, the charge status of the electric vehicle **10**.

(56) In an exemplary embodiment, the monitoring, with the system controller **34A** in communication with the electrical system **52** within the electric vehicle **10**, information related to a charge status of the electric vehicle **10** at block **204** further includes monitoring, with the system controller **34**, at least one of, moving to block **208**, a level of charge **74A** of the electrical system **52** within the electric vehicle **10**, and, moving to block **210**, a time left until the charging of the electric vehicle is complete. The monitoring, with the system controller **34A**, a time left **74B** until the charging of the electric vehicle **10** is complete further includes calculating a time until one of, moving to block **212**, when the electric vehicle **10** is fully charged, and, moving to block **214**, when the electric vehicle **10** is charged sufficiently to allow the electric vehicle **10** to travel from the charging station **64** to a planned destination.

(57) In another exemplary embodiment, the providing notification **72**, to a third-party **66**, with the system controller **34A**, the charge status of the electric vehicle **10** at block **206** further includes, moving to block **216**, receiving, with the system controller **34A** of the electric vehicle **10**, a charging status query from a third-party **66**, wherein the third party **66** initiates the charging status query based on at least one of, moving to block **218**, in response to manual initiation by the third party **66**, moving to block **220**, automatically based on third party navigation and destination information, and, moving to block **222**, automatically based on battery charge level of the third party **66** and proximity to the charging station **64**, and, moving to block **224**, providing notification **72**, to the third-party **66**, with the system controller **34A**, the charge status of the electric vehicle **10** in response to receiving the charging status query.

(58) In an exemplary embodiment, the providing notification **72**, to a third party **66**, with the system controller **34A**, the charge status of the electric vehicle **10** at block **206** further includes, moving to block **226**, providing a visual representation **72A** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, and receiving, by the third party **66**, the visual representation **72A** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete.

(59) In an exemplary embodiment, the providing a visual representation **72A** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete at block **226** further includes, moving to block **228**, displaying, with an internal/external display system **68** in communication with the system controller **34A** and adapted to project information onto a window surface **70** of the electric vehicle **10**, at least one of, moving to block **230**, the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, moving to block **232**, a graphic representation **72B** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, and, moving to block **234**, a QR Code **72C** including the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the

charging of the electric vehicle **10** is complete.

(60) In an exemplary embodiment, the providing a visual representation **72A** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete at block **226** further includes, moving to block **236**, displaying, with the system controller **34A**, via a display **76** of the electric vehicle **10**, at least one of, moving to block **238**, the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, moving to block **240**, a graphic representation **72B** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, and, moving to block **242**, a QR Code **72C** including the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete.

(61) In an exemplary embodiment, the providing a visual representation **72A** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete at block **226** further includes, moving to block **244**, displaying, with the system controller **34A**, via a display screen **77** of the charging station **64**, at least one of, moving to block **246**, the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, moving to block **248**, a graphic representation **72B** of the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete, and, moving to block **250**, a QR Code **72C** including the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete.

(62) In another exemplary embodiment, the providing notification, to a third party **66**, with the system controller **34A**, the charge status of the electric vehicle **10** at block **224** further includes, moving to block **252**, uploading, with the system controller **34A**, the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete to a cloud-based charging station network **65**, moving to block **254**, accessing, by the third party **66**, the cloud-based charging station network **65**, and, moving to block **256**, receiving, by the third party **66**, from the cloud-based charging station network **65**, the current level of charge **74A** of the electric vehicle **10** and the time left **74B** until the charging of the electric vehicle **10** is complete.

(63) In still another exemplary embodiment, the providing notification **72**, to a third party **66**, with the system controller **34A**, the charge status of the electric vehicle **10** at block **224** further includes, moving to block **258**, sending, with the system controller **34A**, notification **72** to the third party **66** that the charging station **64** to which the electric vehicle **10** is connected has malfunctioned.

(64) A system **11** and method **200** of the current disclosure offers the advantage of providing information related to a charge status of an electric vehicle when the electric vehicle is connected to a charging station to a third party, whereby the third party can strategically select a charging station that minimizes the time spent waiting for such charging station.

(65) The description of the present disclosure is merely exemplary in nature and variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure.

Claims

1. A method of providing information related to a charge status of an electric vehicle when the electric vehicle is connected to a charging station, comprising: monitoring, with a system controller in communication with an electrical system within the electric vehicle, information related to a charge status of the electric vehicle; and providing notification, to a third-party, with the system controller, the charge status of the electric vehicle, wherein the third-party is a third-party electric vehicle or a user of the third-party electric vehicle, and providing notification includes: displaying,

with a hybrid augmented reality head up display (HUD) in communication with the system controller and adapted to project information onto a window surface of the electric vehicle that is visible to both occupants within the electric vehicle and persons outside of the electric vehicle, at least one of: a current level of charge of the electric vehicle and a time left until the charging of the electric vehicle is complete; a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

2. The method of claim 1, wherein the monitoring, with the system controller in communication with the electrical system within the electric vehicle, information related to a charge status of the electric vehicle further includes monitoring, with the system controller, at least one of: a level of charge of the electrical system within the electric vehicle; and a time left until the charging of the electric vehicle is complete.

3. The method of claim 2, wherein the monitoring, with the system controller, a time left until the charging of the electric vehicle is complete further includes calculating a pre-determined threshold that is based on at least one of: a time until the electric vehicle is fully charged; and a time until the electric vehicle is charged sufficiently to allow the electric vehicle to travel from the charging station to a planned destination, wherein the method includes: receiving, with the system controller, information related to the planned destination from at least one of: input from a user of the electric vehicle; a personal device of a user of the electric vehicle; or a navigation system within the electric vehicle; or when information related to the planned destination is not available: receiving, with the system controller, from a vehicle charging station infrastructure network, information related to a distance to a next available charging station; and designating the next available charging station as the planned destination.

4. The method of claim 3, wherein the providing notification, to a third-party, with the system controller, the charge status of the electric vehicle further includes: receiving, with the system controller of the electric vehicle, a charging status query from a third-party, wherein the third party initiates the charging status query based on at least one of: in response to manual initiation by the third party; automatically based on third party navigation and destination information; and automatically based on battery charge level of the third party and proximity to the charging station; and providing notification, to the third-party, with the system controller, the charge status of the electric vehicle in response to receiving the charging status query.

5. The method of claim 3, wherein the providing notification, to a third party, with the system controller, the charge status of the electric vehicle further includes: providing a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; and receiving, by the third party, the visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

6. The method of claim 5, wherein the providing a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete further includes displaying, with the system controller, via a display of the electric vehicle, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

7. The method of claim 5, wherein the providing a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete further includes displaying, with the system controller, via a display screen of the charging station, at least one of: the current level of charge of the electric vehicle and the time left until the charging

of the electric vehicle is complete; a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

8. The method of claim 3, wherein the providing notification, to a third party, with the system controller, the charge status of the electric vehicle further includes: uploading, with the system controller, the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete to a cloud-based charging station network; accessing, by the third party, the cloud-based charging station network; and receiving, by the third party, from the cloud-based charging station network, the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

9. The method of claim 8, wherein the providing notification, to a third party, with the system controller, the charge status of the electric vehicle further includes sending, with the system controller, notification to the third party that the charging station to which the electric vehicle is connected has malfunctioned.

10. A system for providing information related to a charge status of an electric vehicle when the electric vehicle is connected to a charging station, comprising: a system controller in communication with an electrical system within the electric vehicle and adapted to: monitor information related to a charge status of the electric vehicle; and provide notification, to a third-party, of the charge status of the electric vehicle, wherein the third-party is a third-party electric vehicle or a user of the third-party electric vehicle, wherein the system controller is further adapted to display, with a hybrid augmented reality head up display (HUD) in communication with the system controller and adapted to project information onto a window surface of the electric vehicle that is visible to both occupants within the electric vehicle and persons outside of the electric vehicle, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

11. The system of claim 10, wherein when monitoring information related to a charge status of the electric vehicle, the system controller is further adapted to monitor a level of charge of the electrical system within the electric vehicle, and a time left until the charging of the electric vehicle is complete.

12. The system of claim 11, wherein when monitoring a time left until the charging of the electric vehicle is complete the system controller is further adapted to calculate a pre-determined threshold that is based on at least one of: a time until the electric vehicle is fully charged; and a time until the electric vehicle is charged sufficiently to allow the electric vehicle to travel from the charging station to a planned destination, wherein the system controller is further adapted to: receive information related to the planned destination from at least one of: input from a user of the electric vehicle; a personal device of a user of the electric vehicle; or a navigation system within the electric vehicle; or when information related to the planned destination is not available: receive, from a vehicle charging station infrastructure network, information related to a distance to a next available charging station; and designate the next available charging station as the planned destination.

13. The system of claim 12, wherein when providing notification, to a third-party, the charge status of the electric vehicle, the system controller is further adapted to: receive a charging status query from a third-party, wherein the third party initiates the charging status query based on at least one of: in response to manual initiation by the third party; automatically, based on third party navigation and destination information; and automatically, based on battery charge level of the third party and proximity to the charging station; and provide notification, to the third-party, the

charge status of the electric vehicle in response to receiving the charging status query.

14. The system of claim 12, wherein when providing notification, to a third party, the charge status of the electric vehicle, the system controller is further adapted to provide a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, wherein, the third party receives the visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

15. The system of claim 14, wherein, when providing a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, the system controller is further adapted to display, with a display of the electric vehicle, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

16. The system of claim 14, wherein, when providing a visual representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete, the system controller is further adapted to display, with a display screen of the charging station, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete.

17. The system of claim 12, wherein, when providing notification, to a third party, the charge status of the electric vehicle, the system controller is further adapted to: upload the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete to a cloud-based charging station network, wherein the cloud-based charging station is adapted to allow access by the third party, and the third party receives the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete from the cloud-based charging station network; and send notification to the third party that the charging station to which the electric vehicle is connected has malfunctioned.

18. An electric vehicle equipped with a system for providing information related to a charge status of the electric vehicle when the electric vehicle is connected to a charging station, the system comprising: a system controller in communication with an electrical system within the electric vehicle and adapted to: monitor a level of charge of the electrical system within the electric vehicle; monitor a time left until the charging of the electric vehicle is complete by calculating at least one of a time until the electric vehicle is fully charged, and a time until the electric vehicle is charged sufficiently to allow the electric vehicle to travel from the charging station to a planned destination, wherein the system controller is further adapted to: receive information related to the planned destination from at least one of input from a user of the electric vehicle, a personal device of a user of the electric vehicle, or a navigation system within the electric vehicle; or when information related to the planned destination is not available, receive, from a vehicle charging station infrastructure network, information related to a distance to a next available charging station, and designate the next available charging station as the planned destination; receive a charging status query from a third-party, wherein the third party initiates the charging status query based on at least one of: in response to manual initiation by the third party; automatically, based on third party navigation and destination information; and automatically, based on battery charge level of the third party and proximity to the charging station; provide notification, to a third-party, the charge status of the electric vehicle in response to receiving the charging status query by at least one of: displaying, with a hybrid augmented reality head up display (HUD) in communication with the

system controller and adapted to project information onto a window surface of the electric vehicle that is visible to both occupants within the electric vehicle and persons outside of the electric vehicle, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; displaying, with a display of the electric vehicle, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; displaying, with a display screen of the charging station, at least one of: the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; a graphic representation of the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; and a QR Code including the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete; and uploading the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete to a cloud-based charging station network, wherein the cloud-based charging station is adapted to allow access by the third party, and the third party receives the current level of charge of the electric vehicle and the time left until the charging of the electric vehicle is complete from the cloud-based charging station network; and sending notification to the third party that the charging station to which the electric vehicle is connected has malfunctioned.
